ShopSafe: Artificial Intelligence-Powered Theft Prevention Using Multiscale Analysis

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Abstract—Shoplifting has become a growing problem for businesses, large and small, whether they are standalone stores or major retail stores. It is usually practiced by people hiding items in the clothes or bags they carry away without paying at the premises. However, no automated approach to this problem is available yet. Therefore, we proposed an artificial intelligence monitoring system that autonomously recognizes shoplifting incidents in retail environments by identifying object-stealing actions in video sequences to address this issue. When such events are detected, the system generates a notification. Our approach proposes a multiscale parallel feature aggregation network to extract salient features in different video sequences. Then, a long short-term memory model was combined to categorize activities and detect shoplifting events. This work makes a significant contribution by optimizing the recurrent learning classifier parameters in relation to sequence length and batch size. We also made available a novel shoplifting dataset, created in a lab-style environment and containing synthesized video clips of normal and shoplifting activities. The proposed system obtained 91.48% accuracy, surpassing existing methods and highlighting its potential as a robust and effective solution for shoplifting detection.

Keywords: human activity, convolutional neural network, multiscale parallel feature aggregation, spatiotemporal information, long short-term memory

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1. INTRODUCTION

Theft of products has become a growing concern for megastores and small retailers. This type of theft is called shoplifting. The people involved in this activity are called shoplifters. Such people hide the stolen goods in bags and clothes, so they do not get caught and do not have to pay for them. Shoplifting results in a financial loss for retailers and undermines the shopping experience entirely [5, 23]. Shoplifting is the act of taking products from retail stores with the intent to avoid paying for them, typically to remain undetected. This is a challenge that cannot be overcome by retailers, and its losses are increasing at a rapid rate, posing a threat to the sustainability and profitability of retailers.

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Shoplifting is a suspicious act that is the opposite of what a typical customer would do. Examples of common activities in retailing include walking, viewing products, billing, and interacting with store staff. On the other hand, shoplifting activities involve stealing items, such as clothes or bags, without being detected by security guards [21, 25]. Generally, such activities are covert and, hence, cannot be easily detected using conventional surveillance practices. Most current surveillance systems are based on closed-circuit television (CCTV) infrastructure. Human operators in such a setup monitor a live video feed to identify suspicious activities. However, these systems are also limited by human attention and the amount of information that multiple cameras can capture. These limitations have rendered manual surveillance ineffective [4, 10, 11, 20, 22, 24], resulting in missed observations and increased response latency. There are some weaknesses in the current conventional surveillance systems. This involved manual monitoring of video frames, requir-

ing individuals to be fully attentive. Owing to heavy mental and physical stress, some key frames are sometimes lost, a fact that augments security risks. Considering this problem, the article introduces an automated and cost-efficient system that identifies malicious practices in real-time by inspecting each frame and sending an alert to the security staff. The objective is to create a smart human activity recognition (HAR) system that understands and categorizes suspicious activities that are associated with shoplifting using a live video feed. The system suggested utilizes recent deep learning (DL) and computer vision technologies to promote the effectiveness and precision of surveillance activities. Shoplifting has never posed greater economic and operational challenges to individuals, which is why this study is motivated to investigate the issue.

Theft, fraud, and administrative errors are components of shoplifting, which are key contributors to inventory shrinkage [20]. Retailers tend to spend a significant amount of money on surveillance equipment and manpower to contain these losses. However, such measures can only be as effective as manual monitoring and intervention can be. The emergence of machine learning (ML) technologies has constituted the latent to transform surveillance practices. Models constructed using DL, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have been found to be exceptionally effective in tasks like pattern and sequence recognition in visual information. These technologies can be incorporated into surveillance systems, resulting in a more effective solution to identify real-time shoplifting cases and eliminate the need for numerous manual reviews. Our proposal is a new system for detecting automatic product theft (shoplifting). The contributions of the research are as follows.

• Proposing an artificial intelligence-based HAR system: this system detects shoplifting behaviors in retail environments precisely using MPFAN and LSTM.

• Developing a high-quality dataset: as no existing dataset is available to train our system, we create a synthesized dataset consisting of normal and suspicious activities. Various environments allow the system to generalize very well in different retail environments.

• Extensive parameter analysis: the paper performs an extensive study of several critical parameters, including sequence length and batch size, to provide useful information about the factors that influence the accuracy and efficiency of the HAR model.

• Real-time detection capability: this system proposes a process to detect human activities from live video streams and give immediate alerts of theft incidents in a shop to security personnel.

This strategy is used to overcome the inefficiencies of conventional CCTV-based programs, offering a scalable, proficient, and accurate solution to the shoplifting issue. Since the DL models are integrated, it is a real-time process to detect and classify suspicious activities, resulting in a smarter and safer retail environment. The study contributes to the bigger objective of applying AI to the welfare of the population, including the safety and efficiency of retail operations.

The remainder of the article, which includes Section 2, presents a literature review of existing HAR systems, detailing their methodologies, applications, and limitations. Section 3 provides the information regarding the proposed methodology. It contains system architecture, dataset preparation, methods of feature extraction, and a recursive classifier. There are also discussions on design choices and critical design aspects. Section 4 provides an explanation of the experimental environment, the characteristics of the dataset, performance criteria, and performance indicators. The proposed system is supported by evidence of its efficiency, which can be demonstrated through experimental results. The research findings and implications are presented in Section 5, which also outlines areas for further research to enhance the scalability and flexibility of the system.

2. RELATED WORK

Today, HAR systems are a widely discussed field due to their diverse applications in surveillance, healthcare, entertainment, and other fields. The systems are used to ensure intelligent automation, safety improvements, and an enhanced user experience, as they can help machines learn human behavior. Relations and activities are interconnected and vary in complexity and scope.

As was noted, there are similarities in actions and activities, but these are different [18]. Actions are brief and quick movements that an individual undertakes, such as the movement of a hand, sitting, or bending, which often take a few seconds. Nonetheless, an activity is a sequence of actions of an individual or a group of individuals that represent more sophisticated and protracted behaviors. As an illustration, an activity may imply committing an assault, shaking hands, or leaving an unattended object in a public place. Ladjailia et al. [14] developed a computer vision-based HAR system. This system entails cognition, monitoring and surveillance. Recent research has focused on the detection and recognition of human behaviors through computer vision with markerless techniques, as well as noninvasive techniques such as intelligent video cameras in natural environments. The field of research took a step further, abandoning traditional feature-based, manual methods, and began employing state-of-the-art DL technologies to handle sophisticated data and real-life variability.

2.1. Handcrafted Feature-Based Methods

The current HAR models utilize manually created features to capture actions. Such features were utilized by Nguyen and Ly [16] to construct an activity recognition model, in which they aimed to identify the movement of objects using a differencing algorithm. The descriptors employed in the extraction of the features in this field of investigation include oriented gradients histogram (HOG), optical flow histogram (HOF), and motion boundary histogram (MBH). These features are then extracted and trained in a support vector data descriptor (SVDD) to differentiate between various behaviors. On the same note, Kushwaha et al. [13] employ HOG gradient vectors along with optical flow motion vectors and an SVM model to detect human activity. Abualigah et al. [1] present optimization techniques to enhance predictive modeling by cutting down the number of input vectors to enhance efficiency and performance. As an example, Sanal and Bhavani [19] used a genetic algorithm to optimize the selection of features, the reduction of the dimensions of the inputs and the enhancement of the predictive capability. Though such systems have been proven to be reliable in their performance, being based on manually developed features restricts their scalability and flexibility in handling complex data. The handmade feature methods tend to lose subtle spatio-temporal interactions, particularly in complex or contaminated settings.

2.2. Deep Learning-Based Methods

The recent development in DL has essentially tittivated HAR research through the capability of automatic feature extraction and representation learning. The CNN model is widely used in image processing and has remarkable accomplishments [3, 25]. However, two-dimensional CNN models are unable to undertake video processing research. Ji et al. [9] alleviated such a deficit using a three-dimensional CNN in the HAR model. They extract features from both space and time using a 3D-CNN, which captures motion data across consecutive frames. Yue-Hei Ng et al. [15] amalgamate a recurrent neural network (RNN) model with CNN to increase the classification efficiency of video frames.

The proposed approach appears more effective, although it still requires some advancements. Donahue et al. [7] developed a method using a neural network called LSTM to recognize video movements. This method is good at learning from both the space (what is happening in the scene) and the time (how things change over time) in the video. They combine this approach with advanced image recognition techniques, helping it to understand the details in each video frame. Noori et al. [17] developed a system that recognizes human activities by looking at body posture and movement. They tracked important points on the body across different video frames to capture the person's movement. They used an advanced DL model to improve the method's performance to identify actions, and by adding movement data, they achieved strong results in activity recognition. Sultani et al. [23] proposed a model to identify unusual or inconsistent actions in the real world using a method that learns from imperfectly labeled examples. Their method assists the system in recognizing patterns even when some of the training data is not fully accurate or clear. They also determined that the deep anomaly ranking model easily predicts higher anomaly scores to separate abnormal videos.

Recent research has focused on hybridizing CNN and LSTM models to overcome these challenges. Asadi-Aghbolaghi et al. [6] investigated HAR using the DL model. In their study, they explore motion and temporal-based descriptors for motion detection. They highlight the success of 3D models in considering temporal patterns across long sequences, making them suitable for capturing temporal relationships. Subsequently, LSTM models are suitable for such types of tasks. Jayaswal and Dixit [8] developed a system to detect unusual actions in real-time scenarios using an advanced DL model, such as Xception, to identify important details. Then, it applies another model to classify these events as normal or unusual.

Despite these advancements in the DL models, several challenges persist, such as computational overhead, handling of real-world variability, and long-term temporal dependencies.

2.3. Research Gap

Current HAR research requires features of both handcrafted feature-based and DL-based approaches. Current handcrafted feature-based HAR systems heavily depend on manually designed descriptors that are challenged in extracting information from video sequences due to the loss of global data structure through interpolation lattices. However, these methods largely fail to capture intricate patterns in complex datasets. On the other hand, DL-based systems for recognizing human activities that use the CNN model automatically learn important patterns from videos. They do this by directly analyzing the key details in the video sequences. Therefore, CNNs are undoubtedly very effective HAR methods thanks to their ability to retain important information with learned features. However, most researchers who have leveraged this approach have only used traditional ML classifiers coupled with CNNs, which do not allow them to model long-term temporal relationships. To remedy this shortcoming, the methods utilize recurrent learning models, including recurrent neural networks (RNNs), which have a notion of memory to remember and process information over time using memory components. RNNs overcome CNNs by capturing temporal dependencies; however, they share the same

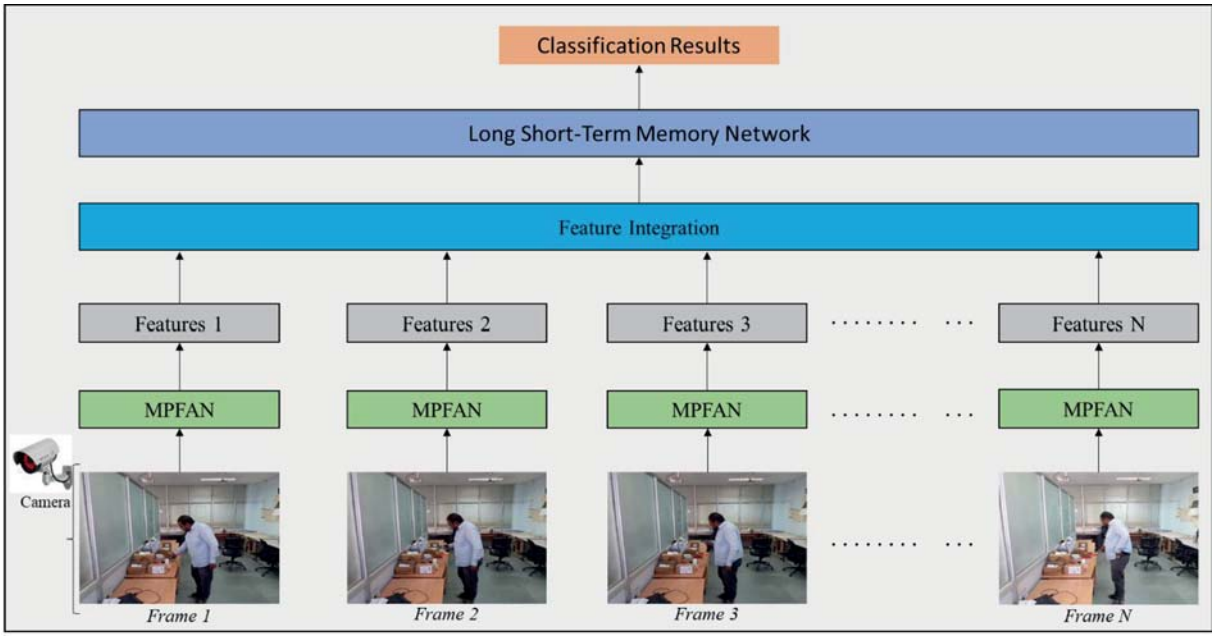


Fig. 1. Pipeline for shoplifting action detection.

problem of long-term dependencies, which can lead to gradient issues. We introduce the (long short-term memory) LSTM networks to overcome these limitations and enhance RNN networks. LSTMs enhance time domain learning and can mitigate the gradient vanishing and explosion problems when trained for an extended period with large datasets. In this study, we propose an approach that combines the MPFAN for spatial feature extraction and LSTM networks for temporal relationship modeling to build a hybrid system that successfully learns both the spatial and temporal information of human activities. In this work, we propose a hybrid model proven superior to previous methods in accurately recognizing human actions, especially in complex settings such as human stealing actions. Moreover, performance optimization in this research is achieved by fine-tuning sequence length and batch size. It provides a more comprehensive understanding of the behavior of the model. The amalgamation of MPFAN and LSTM provides a robust solution to the challenges facing HAR, outperforming current HAR systems in terms of accuracy and scalability improvements.

3. PROPOSED METHODOLOGY

In this paper, we introduce a state-of-the-art system of HAR that can identify the act of shoplifting in real-time situations. The steps in the system are image capturing, feature extraction and classification as depicted in Fig. 1.

The first step involves capturing a real-time video frame by the CCTV camera in the input scene, which is then continuously fed into the system. These frames are then transferred to RGB so that the significant visual data of the scene can be saved. These frames are in RGB format and are forwarded to the feature extraction module for processing. During this phase, the data is prepared for processing at the next level, retaining only relevant information.

The second step involves concentration on the spatial characteristics of the frame captured by means of an MPFAN model. The model excels in acquisition and pattern recognition of image data. The model operates on the video frame separately, retrieving features as visual information in a compact form of representation. One vector is generated by a sequence of these features, with a length of N, which corresponds to the sequence length. Such frames enable the model to learn temporal relational motion changes over time, which is critical for capturing activities like shoplifting. Extracting meaningful features from these sequential frames in real time is critical for discriminating between different human activities.

The final step is classification, which utilities the extracted features to train an LSTM network. The LSTM model is good at learning and remembering long-term dependencies in sequential data. Then, the human activity is classified in the video segment by the LSTM network using normal behavior and the possibility of shoplifting events. The classification enables the system to make real-time decisions on suspicious activities and provides timely alerts for action. To manage large video streams, the system cuts the input video into smaller substreams so that each substream is of SL length. The classification module operates independently on each substream. Each sequence is then

fed to the action recognition module to extract features and to the LSTM network for classification of the associated substream's activities. The process for the substream task is repetitive until the complete video sequence has been processed so that no action is missed.

Feature extraction and classification are two crucial phases in developing a realistic HAR system, as implemented in the proposed algorithm. The following subsections discuss the DL approaches to structuring an intelligent surveillance system.

3.1. Multiscale Parallel Feature Aggregation Network (MPFAN)

A multibranch CNN model incorporating a multiscale feature extraction architecture is proposed to learn multiscale features effectively and achieve robust performance, as shown in Fig. 2. It has an input layer with an image shape of 120 × 120 × 3. Input is processed independently in two parallel branches using convolutional layers (conv1 and conv2) with different kernel sizes and strides. With this configuration, the network can extract various features from the input images. Each branch consists of convolutional blocks, which are then completed with max-pooling layers to reduce spatial dimensions while retaining significant hierarchical information. One of the important features of the model is the use of concatenation layers (feat1_out, feat3_out), which combine feature maps of different branches. It enables the network to retain and integrate information from multiple scales, ensuring the preservation of both global and local spatial patterns. To enjoy the greater accuracy of Thompson sampling in the case of stochastic bandits, we propose a multibranch method that supports more complex input data, utilizing both empirical estimations of values and Thompson sampling. Once the feature extraction process is over, the network moves to the so-called fully connected layers, where it undergoes a high-dimensional transformation and is then classified. The initial stage involves flattening the feature maps to convert the 3D tensors into 1D feature vectors. Next, we trained two deep layers, each consisting of up to 2048 neurons, to learn from more complex structures than just lines and planes in the data. These obtained features are then fed through a second dense layer comprising 512 neurons to further refine them.

To ensure that the model oversimplifies well on test data, dropout layers have been applied after each dense layer to prevent the model from overfitting. The last dense layer consists of a single neuron with either a linear or a sigmoid activation function, depending on whether we are performing a binary classification. With its multibranch design, layered structure, feature concatenation, and regularization techniques, the network is naturally suitable for tasks defined in high-dimensional input spaces and requires precise feature extraction and classification.

3.2. Long Short-Term Memory (LSTM)

The LSTM model [2, 12] demonstrated significant effectiveness in addressing problems involving temporal sequences, such as machine translation, video tagging, image captioning, speech recognition, etc. LSTM is used to handle long-term dependencies more effectively. An LSTM model contains an Input Gate (I), a Forget Gate (Fo), an Output Gate (O), and a Memory Cell (C). These gates enable the LSTM to learn patterns and retain or store information throughout any given time frame intervals.

T represents the time when information is processed and propagated by the combination of the input data point (X_T), the previous hidden state ($H_{(T-1)}$), the current hidden state (H_T), and the cell state (C_T). The LSTM achieves this by a sequence of updates to its parameters, which are performed in subsequent phases. To begin with, the cell state is utilized as a memory in an LSTM network, which saves valuable information within the sequence. The steps followed in the process of regulating or retaining information or adding new information to the cell state are as follows: the usual formulas are Eqs. (1)–(3)

$$Fo_T = \sigma\left[W_F.\left(H_{T-1}, X_T\right) + B_F\right], \tag{1}$$

$$I_T = \sigma\left[W_I.\left(H_{T-1}, X_T\right) + B_I\right], \tag{2}$$

$$\tilde{C}_T = \tanh\left[W_C.\left(H_{T-1}, X_T\right) + B_C\right]. \tag{3}$$

In the second phase, the earlier cell state is changed to the new cell state, as illustrated in Eq. (4)

$$C_T = F_T \cdot C_{(T-1)} + I_T \cdot \tilde{C}_T. \tag{4}$$

The third phase generates an output filtered based on the cell state, as demonstrated in Eqs. (5) and (6)

$$O_T = \sigma\left[W_O \cdot \left(H_{(T-1)}, X_T\right) + B_O\right]. \tag{5}$$

$$H_T = \tanh\left(C_T\right) \cdot O_T. \tag{6}$$

In this context, I_T (input gate), O_T (output gate), F_T (forget gate), and W_I, W_O, W_F, and W_C are the weight matrices. Meanwhile, B_I, B_O, B_F, and B_C represent the corresponding biases and the C_T (current cell state), $C_{(T-1)}$ (previous cell state), and $\tilde{C}_T$ (candidate cell state) represents different states. Sigmoid (σ) activation and hyperbolic tangent function (tanh) are essential components of deep recurrent networks.

The output of an LSTM is calculated using Eq. (7), where the output from the previous step is used as input for the current step

$$H_T = F_W\left(X_T, H_{(T-1)}\right). \tag{7}$$

In the final phase, the softmax function is used to compute the likelihood scores for each class, enabling class prediction.

The evaluation results of the proposed method, using two different shoplifting datasets, are presented in the next section. Additionally, a comparison is con-

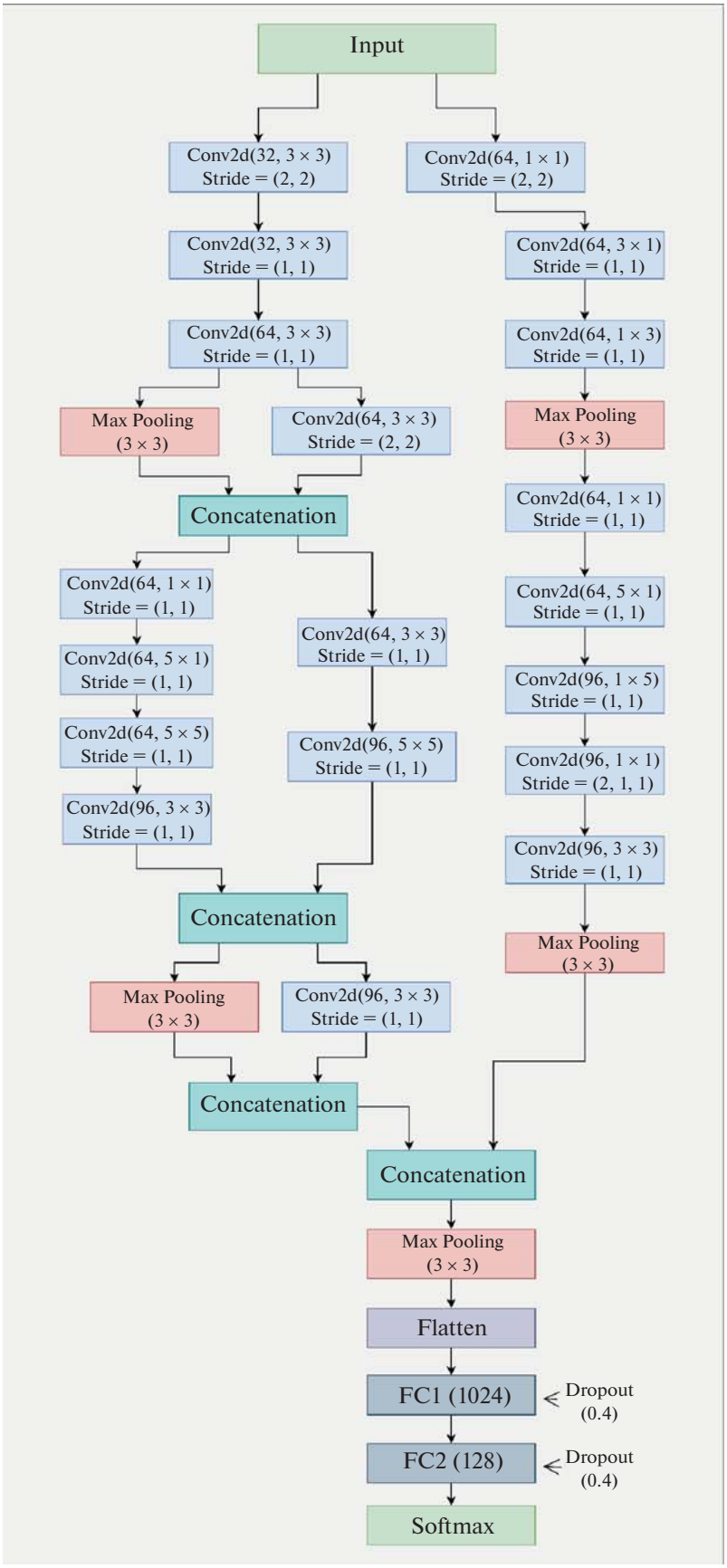


Fig. 2. Multibranch CNN for multiscale feature extraction.

ducted on the system's effectiveness in detecting theft activities under various modeling parameter configurations.

4. EXPERIMENTS AND RESULT ANALYSIS

The experiments were performed on a Core i7-13700 CPU, 24 GB of RAM, and an NVIDIA GeForce 4060 GPU with 8 GB of memory. The software versions for the GPU used are CUDNN 7.6.3 and CUDA 10.0.130. The experimentation involved tuning parameters to determine the effects of the setups on the model's predictive behavior under different conditions. The main parameters that were tuned in the proposed model are the sequence length (SL) and the batch size. The entire action is characterized by the number of frames in the video clips, which determines the SL parameter. Significant experiments were conducted to determine various SL values. The batch size regulates the quantity of samples given to the model prior to modifying its internal parameters, which is crucial for maintaining stable network performance. Several batch sizes were experimented with (i.e., 4, 8, 16, and 32) and the results of each were compared. Additionally, a parameter of patience was applied to determine the stopping criterion during training. This parameter was set to 50 in all experiments, indicating that training would automatically cease if validation losses were not decreasing in 50 epocs, thereby preventing the model from overfitting. Based on these deliberations, the experiment was conducted with different sequence length (SL) parameters using multiple shoplifting datasets with varying batch sizes. Then, the results were analyzed to identify the most effective configuration.

4.1. Datasets

The proposed model was evaluated using two datasets: One dataset is the first from the UCF repository [23], and the second dataset is synthesized specifically for this research. The UCF Crime dataset comprises 128 h of uncut crime videos with 13 realistic and 1900 irregularities. These represent various real-world events, such as theft, robbery, abuse, arson, shooting, vandalism, shoplifting, explosions, fighting, burglary, and arrest. We separated normal and shopper video clips using 5-second continuous clips from a large pool of CCTV footage and then prepared these clips by trimming human activity for this study.

After trimming, the resulting set of shoplifting video clips is referred to as the UCF dataset in this manuscript. The dataset comprises classes for stealing and normal actions, with a total of 126 clips. Out of these, 82 clips represent normal actions, and 44 clips depict shoplifting incidents. Each clip was captured at a resolution of 320 × 240 pixels and at 30 frames per second. Since the dataset was small and deep neural networks require large, labeled datasets for effective training, we applied flip augmentation to expand the dataset. By making the dataset invariant to left-right rotation, this augmentation effectively doubled the number of clips available for use in the dataset. After augmentation, our dataset comprised 252 clips, 164 of which represented normal actions and 88 represented shoplifting events. The 174 clips were trained on this network and then tested on 78. Table 1 has statistics of clips in the UCF dataset. Figure 3 shows sample frames for the shoplifting class and the normal class of the dataset.

Table 1. Distribution of clips and frames in the dataset

Dispersal	Class	UCF	Synthesized
Test	Normal	54	48
	Shoplifting	24	46
Train	Normal	110	128
	Shoplifting	64	128
Total clips		252	350

Shoplifting events are an example shown in the first row of Fig. 3. An individual is seen concealing an item, indicating a potential theft. The second row represents frames of normal activities. (Walking, watching, examining products in a shop area). By analyzing each clip in that dataset in detail, we found some inconsistencies in the clips that would affect the performance of our recognition model. These issues are summarized as follows.

• Inter-object and self-occlusion. This refers to when one object obstructs another, like in the case of a person's stealing action being blocked by stationary or moving objects (people, baggage, etc.). Self-occlusion occurs when the subject's body parts or clothing cover the stealing action, making it difficult to perceive the activity.

• Viewpoint variation. Viewpoint variations appear in the dataset; some clips are shot too close to the subject, and some are far from it. Occlusion may occur in clips filmed at a close range, which naturally obstructs the human body and the action. On the other hand, clips taken further away provide a wider view but lose detail, as the viewer may be unable to distinguish what action is taking place. Thus, the precision of the recognition process is affected by the varying distances from the camera to the subject.

• Lighting variations. Visual recognition tasks depend substantially on lighting because changing lighting conditions impact object visibility and appearance in the scene. With the UCF dataset, the shoplifting clips feature different lighting conditions, like clips that are underexposed or overexposed frames. To make matters worse, these variations impact the clarity of human actions and hinder the model's ineffective identification and classification of activities across various lighting scenarios.

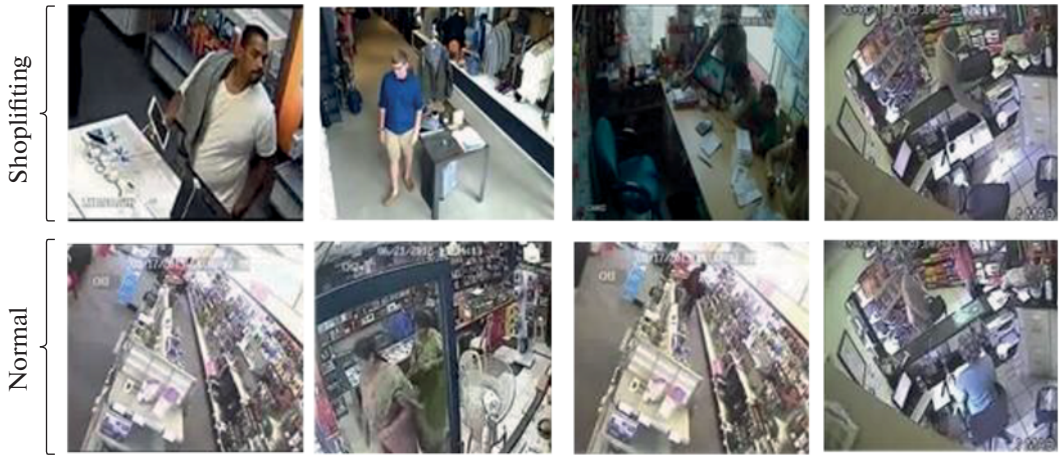


Fig. 3. Image instances from the UCF dataset.

The UCF shoplifting datasets, which contain inconsistent data, limit intuitive human action recognition. We created a synthesized video dataset captured in a controlled laboratory setting to circumvent these limitations. Given the authenticity and real-world applicability of our dataset, we aimed to replicate shoplifting attacks as closely as possible to the real world. This research has utilized a novel data set that explicitly targets modeling shoplifting activities, especially stealing and hiding within the shop. A 32-megapixel mobile camera was used to record the videos at 640 × 480 pixels, and each clip lasts 10 s. Similar to the UCF dataset, the synthesized dataset is divided into two categories: shoplifting and normal classes. A normal class is everything we do in our everyday lives, e.g., walking, talking to others, and looking at things.

On the other hand, the shoplifting class demonstrates actions such as hiding retail products under clothing or in a bag. The synthesized dataset contains 175 video clips, out of which 88 are normal actions, and 87 are shoplifting. A flip augmentation was applied to improve the quality and make this dataset robust. However, the process then doubles the number of clips to 350. After updating, the shoplifting class comprises 176 clips, and the normal class comprises 174. Then, the dataset is split into training and testing sets. This result is achieved by training the model on 256 clips, with the remainder used for testing. Table 1 describes the shoplifting dataset, including clips distributed across both classes.

Figure 4 displays sample frames from the self-synthesized shoplifting dataset. The first row showcases frames of shoplifting events, where individuals are seen engaging in stealing actions. The second row contrasts this by illustrating normal human behaviors commonly observed in stores, such as walking or interacting with products.

4.2. Experimental Outcomes

We used two key parameters for the modeling process: sequence length (SL) and batch size, which are essential for this to work. Various configurations of the proposed model were evaluated with a comprehensive set of experiments. The sequence length (SL) of this experiment was set to 125 and 290 video clips for the shoplifting and synthesized datasets, respectively. The impact of sequence length on model performance was further tested for alternative SL values. The third set of sequences involved using every second frame (SL/2), every third frame (SL/3), and every fourth frame (SL/4). We also explored various sequences to determine the optimal sequence length for accurately recognizing human actions. At the same time, it must be reasonable in terms of performance and computational cost.

Developing a human action recognition system depends on the feature extraction process. The proposed system uses the MPFAN model to extract meaningful features from video sequences. However, this feature extraction process is computationally expensive and time-consuming, particularly as the sequence length increases. Figure 5 illustrates the time required for feature extraction. The model was trained on the UCF and synthesized shoplifting datasets. Upon comparison, we realized that the model trained on the UCF dataset with a sequence length of 125 took a significant amount of time to process the video clips. In contrast, the model trained on the synthesized shoplifting dataset with a sequence length of 290 underwent an even longer feature extraction phase due to the extended sequence length.

These results highlight the trade-off between the sequence length and the time required to extract features, which is crucial to consider when designing an efficient HAR system. The results highlight the need to carefully choose the most suitable sequence length



Fig. 4. Sample frames from the self-synthesized shoplifting dataset.

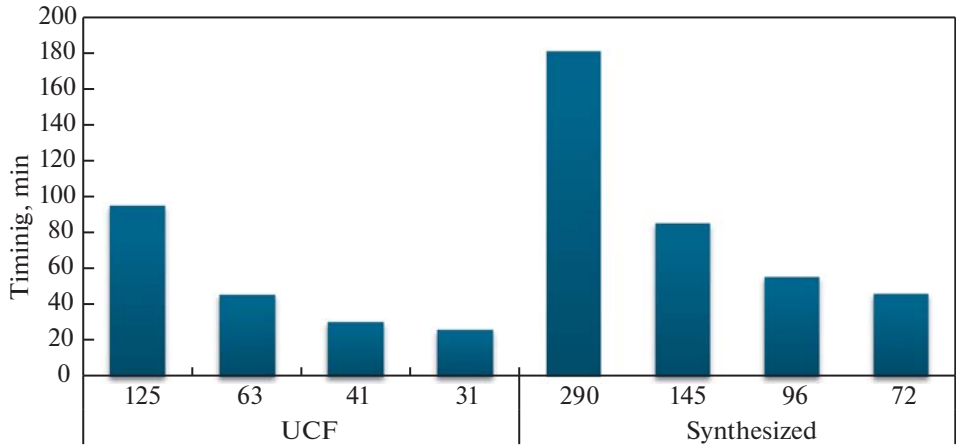


Fig. 5. Feature extraction time for different values of sequence length (SL) for both datasets.

to achieve optimal action recognition while minimizing the system's computational resources.

The next step is to use an LSTM model to classify video sequences into normal activities and shoplifting events. This model classifies video sequences into normal activities and shoplifting events. Since the LSTM model captures temporal dependencies and learns patterns, recognizing actions from video frames is essential. Table 2 presents the results of experiments involving variations in sequence length (SL) and batch size. Here, the models' assessment is done for the different values of SLs. For instance, the model uses a batch size of 32 with an SL of 125 obtained training accuracy (98.98%) and validation accuracy (73.68%). Similarly, a batch size of 16 with an SL of 63 in the model, which gave training and validation accuracy, is 98.56 and 80.76%, respectively. In another experiment, we used a batch size of 4 with an SL of 41 in the model to achieve training and validation accuracy of 99.41 and 80.76%, respectively. With an SL of 31 and the batch size set to 32 in the proposed model, the obtained training and validation accuracy is 99.42 and 79.49%, respectively. Upon analysis, the final model with an SL of 63 and a batch size of 16 produced the highest training and validation accuracy compared to all other SL and batch size configurations.

Next, we examined the synthesized dataset with the different SLs. For the model with a batch size of 16 and an SL of 290, the training accuracy reached 93.33%, while the validation accuracy was 89.23%. For an SL of 145 and a batch size of 8, the model achieved 95.14% training accuracy and 91.48% validation accuracy. Moreover, with an SL of 72 and a batch size of 4, the model attained 100% training accuracy and 90.3% validation accuracy. Finally, when the batch size was 4, the model reached 99.60% training accuracy and 90.3% validation accuracy. The most optimal combination will also lead to almost perfect training accuracy and good validation results.

Table 2. Evaluation results for the UCF and synthesized shoplifting dataset

Batch size	UCF			Synthesized		
	SL	TA (%)	VA (%)	SL	TA (%)	VA (%)
4	125	99.41	68.42	290	86.48	84.71
8		99.28	72.36		89.53	86.04
16		98.56	68.42		93.33	89.23
32		98.98	73.68		86.48	84.97
4	63	100	73.07	145	89.35	84.97
8		99.41	76.92		95.14	**91.48**
16		98.56	**80.76**		88.9	88.1
32		100	78.21		89.05	89.23
4	41	99.41	**80.76**	72	100	90.3
8		98.56	73.08		99.68	88.1
16		97.84	75.64		99.10	89.23
32		97.84	79.49		98.41	87.11
4	31	97.69	75.64	36	99.60	90.3
8		98.84	70.51		98.41	88.3
16		97.69	73.08		98.10	89.23
32		99.42	79.49		95.68	86.04

* TA is training accuracy, * VA is validation accuracy.

4.3. Experimental Analysis

During the training process, an epoch is a unit in which the model's internal parameters are adjusted according to the training data, which assists the learning algorithm in enhancing the model. To measure the effect of varied configurations on training efficiency, we determined the mean epoch timing of our proposed model on UCF and synthesized shoplifting datasets. The hyperparameters have a substantial influence on the mean epoch time for both datasets, as illustrated in Fig. 6.

As we can see, the proposed model requires more time to train each epoch, and this time is proportional to the length of the actual sequence (125 in the UCF dataset and 290 in the synthesized dataset), thus resulting in an increase in total training time. Conversely, a model with shorter sequence lengths (SL/4) or when trained on a generated dataset with an SL of 72, or the UCF dataset with an SL of 30, is less time-consuming to run through an epoch, making it faster to train. For example, with the UCF dataset and a batch size of 4, the model takes 42.39 s per epoch, resulting in approximately 51 min and 37 s of total training time. When the sequence length is reduced to SL/4 and the batch size is set to 8, the average epoch time is reduced to 7.7 s, resulting in a training time of approximately 7 min and 30 s.

The time per epoch of the synthesized dataset is 180 s, which means that the total training time is 403 min and 32 s using the actual sequence length and a batch size of 4. The sequence length reduced to SL/4 with a batch size of 32 yields a mean epoch time of 14.21 s, resulting in a total training time of 17 min and 34 s. We found that training small batches takes longer than training large batches, suggesting that batch size is a crucial factor in determining the training duration.

Finally, the proposed system was tested on both datasets, yielding promising results across various settings. The comparison curve (Accuracy) in Fig. 7 illustrates the model's performance with varying sequence lengths. The SL/SL/2-trained model yielded the best accuracy on both datasets. Particularly, SL/2 showed the best results on both the synthesized and UCF datasets. The SL/3 configuration worked well on the UCF data, and the SL/4 configurations did not increase accuracy by much, yet they were better than SL/3. The model trained on the synthesized dataset had a validation accuracy of 91.48%, which is compared to 80.76% for the UCF dataset. These findings demonstrate that the model worked better on the synthesized data, probably because it was controlled and focused on shoplifting activities.

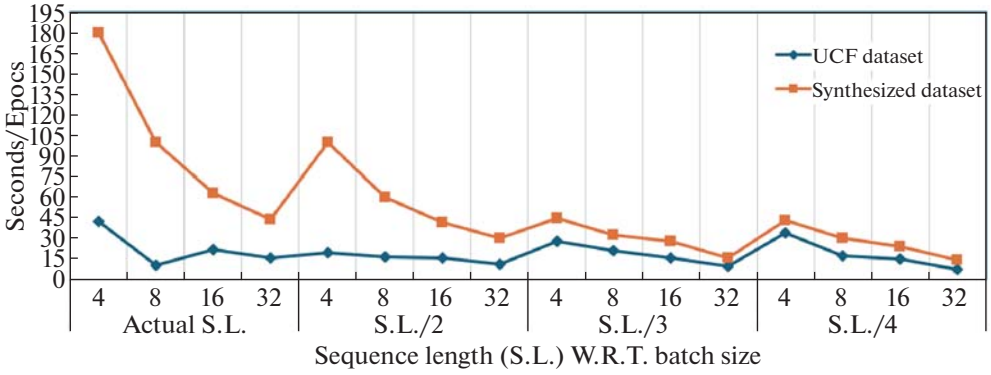


Fig. 6. Comparison of average epoch processing time for both datasets.

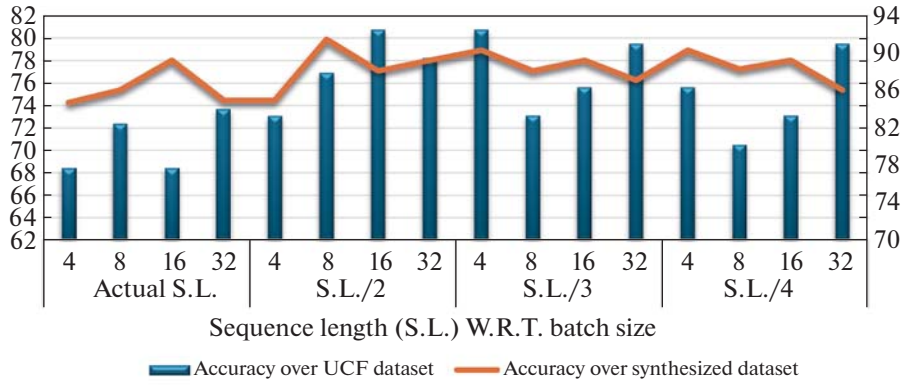


Fig. 7. Comparison of accuracy across different model cases.

4.4. Comparison with Existing Approaches

We compare our proposed model with various existing approaches in the literature, as shown in Table 3. Ansari and Kumar [2] employ a CNN in conjunction with an LSTM network for human activity detection. Ji et al. [9] employ a 3D CNN for HAR to build spatiotemporal relationships in video sequences. On the other hand, Jayaswal and Dixit [8] utilize the Xception network and an RNN classification to recognize video activities, emphasizing the use of temporal context in action recognition. Kurchaniya and Kumar [12] also focus on sequential data.

Unlike these methods, our model more effectively integrates spatial and temporal information by utilizing the deep MPFAN architecture combined with long short-term memory (LSTM) networks. The spatial features extracted from video frames and temporal dependencies over time are combined, enabling the model to accurately capture a more complex set of human actions. We evaluate the proposed model on both UCF and synthesized shoplifting datasets and conclude that it outperforms existing approaches, achieving better accuracy and robustness and providing more precise predictions for the action recognition task. This suggests that combining LSTM networks with MPFAN and incorporating spatial-temporal information offers a significant advantage over conventional methods.

Table 3. Accuracy comparison of models on the synthesized and UCF datasets

Approaches	Accuracy, %	
	UCF	synthesized
Ji et al. [9]	76.92	84.84
Jayaswal and Dixit [8]	78.20	87.23
Kurchaniya and Kumar [12]	79.48	89.36
Ansari and Singh [2]	80.76	89.36
Proposed model	80.76	91.48

5. CONCLUSIONS

This work acknowledges recent progress in deep neural frameworks. It applies the MPFAN architecture for activity recognition coupled with LSTM networks to detect shoplifting activities in a retail environment. The research presents a synthesized shoplifting dataset aimed at reducing merchandise loss caused by theft, focusing on distinguishing between normal and theft-related activities. The deep LSTM network, in conjunction with the proposed model, has great potential for encoding the human dynamic behavior into meaningful actions, which are then classified to distinguish these behaviors. Experimental results indicate that the proposed model is effective, with a sequence length of SL/2, and the system reaches a detection accuracy of approximately 80.76% on the UCF dataset and 91.48% on the synthesized dataset. These results surpass those of existing human action recognition (HAR) systems. The next step of future research could be to extend the shoplifting dataset synthesized as we introduce more specific action classes that are more specific to different stealing behaviors a human would do. In addition, integrating other DL-based HAR techniques may enable the system to characterize human pose semantics better, improving the ability to distinguish between different human actions in video sequences.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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